

SEQUENCE & PROGRESSION

JEE EXERCISES

Exercise J-I

AP & AM BASED

1. There are n A.M's between 3 and 54, such that the 8th mean: $(n-2)^{\text{th}}$ mean :: 3: 5. The value of n is :

यदि 3 और 54 के मध्य n समान्तर माध्य इस प्रकार हैं कि 8 वाँ माध्य: $(n-2)^{\text{th}}$ वाँ माध्य :: 3: 5 हो, तो n का मान है :

- (A) 12 (B) 16 (C) 18 (D) 20

Ans. B

Sol. $\frac{(54-3)}{n+1} = d$; $d = \frac{51}{n+1}$

$$\frac{A_8}{A_{n-2}} = \frac{3}{5} \Rightarrow \frac{3 + 8 \cdot \frac{51}{n+1}}{3 + (n-2) \cdot \frac{51}{n+1}} = \frac{3}{5} \Rightarrow \frac{3n + 3 + 408}{3n + 3 + 51n - 102} = \frac{3}{5}$$

$$\Rightarrow 15n + 2055 = 162n - 297 \Rightarrow 147n = 2352 ; n = 16$$

2. Sum of the series $S = 1^2 - 2^2 + 3^2 - 4^2 + \dots - 2002^2 + 2003^2$ is :

- (A) 2007006 (B) 1005004 (C) 2000506 (D) None of these

श्रेणी $S = 1^2 - 2^2 + 3^2 - 4^2 + \dots - 2002^2 + 2003^2$ का योगफल है :

- (A) 2007006 (B) 1005004 (C) 2000506 (D) इनमें से कोई नहीं

Ans. A

Sol. $S = 1^2 - 2^2 + 3^2 - 4^2 + \dots - 2002^2 + 2003^2 = \sum_{n=1}^{2003} n^2 - 2 [2^2 + 4^2 + \dots + 2002^2]$

$$= \left[\sum_{n=1}^{1001} n^2 \right] = \frac{2003 \times 2004 \times 4007}{6} - \frac{8(1001)(1002)(2003)}{6}$$

$$= \frac{2003 \times 2004}{6} [4007 - 4004] = 2007006 \text{ Ans}$$

3. If $1, \log_9(3^{1-x} + 2), \log_3(4 \cdot 3^x - 1)$ are in A.P. then x equals :

यदि $1, \log_9(3^{1-x} + 2), \log_3(4 \cdot 3^x - 1)$ स.श्रे. में है तो x होगा :

- (A) $\log_3 4$ (B) $1 - \log_3 4$ (C) $1 - \log_4 3$ (D) $\log_4 3$

Ans. B

Sol. $1, \log_9(3^{1-x} + 2), \log_3(4 \cdot 3^x - 1)$ are in A.P.

$$2\log_9(3^{1-x} + 2) = 1 + \log_3(4 \cdot 3^x - 1)$$

$$\log_3(3^{1-x} + 2) = \log_3(4 \cdot 3^x - 1)$$

$$3^{1-x} + 2 = 12 \cdot 3^x - 3$$

$$\frac{3}{3^x} = 12 \cdot 3^x - 5$$

$$\text{Let } 3^x = t$$

$$12t^2 - 5t - 3 = 0$$

$$(3t + 1)(4t - 3) = 0$$

$$t = -\frac{1}{3}, t = \frac{5}{4}$$

$$3^x = -\frac{1}{3}$$

$$3^x = \frac{3}{4}$$

Not possible

$$\log_3 3^x = \log\left(\frac{3}{4}\right)$$

$$x = 1 - \log_3 4$$

4. If the third term of an A.P. is 7 and its 7th term is 2 more than three times of its 3rd term, then sum of its first 20 terms is :

यदि एक समान्तर श्रेणी का तीसरा पद 7 तथा इसका 7वाँ पद इसके तीसरे पद के तीन गुने से 2 अधिक है, तो इसके प्रथम 20 पदों का योग होगा :

(A) 228

(B) 74

(C) 740

(D) 1090

Ans. C

Sol. $T_3 = 7, \quad T_7 = 2 + 3 \times T_3 = 2 + 3 \times 7$

$$T_7 = 2 + 21 = 23$$

$$a + 2d = 7 \quad \dots(1)$$

$$a + 6d = 23 \quad \dots(2)$$

$$(1) - (2) : (2d - 6d) = (7 - 23)$$

$$-4d = -16 \Rightarrow d = 4 \Rightarrow a = -1$$

$$S_{20} = \frac{20}{2} [2 \times (-1) + (20 - 1)4]$$

$$= 10 [-2 + 76]$$

$$= 740$$

5. Let T_r be the r^{th} term of an A.P. whose first term is a and common difference is d . If for some positive integers $m, n, m \neq n, T_m = \frac{1}{n}$ and $T_n = \frac{1}{m}$, then $a - d$ equals :

$$T_m = \frac{1}{n} \text{ and } T_n = \frac{1}{m}, \text{ then } a - d \text{ equals :}$$

माना किसी समान्तर श्रेणी का r वाँ पद T_r है। पहला पद a तथा सार्व अन्तर d है। यदि किन्हीं दो धन पूर्णाकों m तथा n , के लिए ($m \neq n$) यदि $T_m = \frac{1}{n}$ एवं $T_n = \frac{1}{m}$ हो, तो $a - d$ बराबर है :

(A) 0 (B) 1 (C) $1/mn$ (D) $\frac{1}{m} + \frac{1}{n}$

Ans. A

Sol. $T_r = a + (r - 1)d$

$$T_m = \frac{1}{n} = a + (m - 1)d \quad \dots(1)$$

$$T_n = \frac{1}{m} = a + (n - 1)d \quad \dots(2)$$

Solving (1), (2), we get

$$a = \frac{1}{mn} = d$$

$$\Rightarrow a - d = 0$$

6. Let $a_1, a_2, a_3, \dots$ be terms of an A.P. If $\frac{a_1 + a_2 + \dots + a_p}{a_1 + a_2 + \dots + a_q} = \frac{p^2}{q^2}$, $p \neq q$ then $\frac{a_6}{a_{21}}$ equals :

माना $a_1, a_2, a_3, \dots$ एक समान्तर श्रेणी के पद हैं। यदि $\frac{a_1 + a_2 + \dots + a_p}{a_1 + a_2 + \dots + a_q} = \frac{p^2}{q^2}$, $p \neq q$ तो $\frac{a_6}{a_{21}}$ का मान है :

(A) $\frac{7}{2}$ (B) $\frac{2}{7}$ (C) $\frac{11}{41}$ (D) $\frac{41}{11}$

Ans. C

Sol. $\therefore \frac{(S_m)_{IAP}}{(S_n)_{IIAP}} = \frac{m^2}{n^2}$ then $\frac{T_m}{T_n} = \frac{2m-1}{2n-1}$

$$\text{so } \frac{a_6}{a_{21}} = \frac{2 \times 6 - 1}{2 \times 21 - 1} = \frac{11}{41}$$

7. The number of natural numbers < 300 that are divisible by 6 but not by 9 is :

300 से कम प्राकृत संख्याओं की संख्या होगी, जो कि 6 से विभाजित है लेकिन 9 से नहीं :

(A) 49 (B) 37 (C) 33 (D) 16

Ans. C

Sol. Divisible by 6 $\Rightarrow$ Common difference = 6

First term = $a = 6$

$$T_n = a + (n-1)d = 6 + (n-1)6$$

$$6 + (n-1)6 < 300$$

$$1 + n - 1 < 50$$

$$n < 50 \Rightarrow n = 49$$

Divisible by 6, 9 $\Rightarrow$ Common difference = LCM (6, 9) = 18

First term = 18

$$T_m = 18 + (m-1)18 < 300$$

$$18m < 300$$

$$m < 16.66 \Rightarrow m = 16$$

Total terms divisible by 6 but not 9 = $49 - 16 = 33$

8. A person is to count 4500 currency notes. Let a_n denote the number of notes he counts in the n^{th} minute. If $a_1 = a_2 = \dots a_{10} = 150$ and $a_{10}, a_{11}, \dots$ are in AP with common difference of -2, then the time taken by him to count all notes is : **[AIEEE-2010]**

- (A) 24 minutes (B) 34 minutes (C) 125 minutes (D) 135 minutes

वह व्यक्ति को 4500 मुद्रा नोट गिनने हैं। माना $n^{\text{वें}}$ मिनट में वह a_n नोट गिनता है। यदि $a_1 = a_2 = \dots a_{10} = 150$ तथा $a_{10}, a_{11}, \dots$ एक समांतर श्रेणी में है जिसका सार्वअन्तर -2 है, तो उसे सभी नोटों को गिनने में समय लगेगा : **[AIEEE-2010]**

- (A) 24 मिनट (B) 34 मिनट (C) 125 मिनट (D) 135 मिनट

Ans. B

Sol. Total currency = 4500

$$a_1 = a_2 = \dots = a_{10} = 150$$

We count $10 \times 150 = 1500$ currency in 10 minutes

Now, total currency left $4500 - 1500 = 3000$

$$a_{11} = 150 - 2 = 148$$

$$a_{11}, a_{12}, a_{13}, \dots, a_n \dots = 148, 146, 144, \dots$$

$$S_n = \frac{n}{2} [2 \times 148 + (n-1)(-2)]$$

$$3000 = n [148 - n + 1]$$

$$3000 = 149n - n^2$$

$$n^2 - 149n + 3000 = 0$$

$$n = \frac{149 \pm \sqrt{101}}{2}$$

$$n_1 = 125 \text{ not possible} \quad n_2 = 24$$

So, total minutes = $24 + 10 = 34$

9. Let S_1 be the sum of first $2n$ terms of an arithmetic progression. Let S_2 be the sum of first $4n$ terms of the same arithmetic progression. If $(S_2 - S_1)$ is 1000, then the sum of the first $6n$ terms of the arithmetic progression is equal to:

[JEE(Main)Mar.2021]

माना एक समांतर श्रेणी के प्रथम $2n$ पदों का योगफल S_1 है। माना उसी समांतर श्रेणी के प्रथम $4n$ पदों का योगफल S_2 है। यदि $(S_2 - S_1) = 1000$ है, तो इस समांतर श्रेणी के प्रथम $6n$ पदों का योग बराबर है :

[JEE(Main)Mar.2021]

- (A) 7000 (B) 5000 (C) 3000 (D) 1000

Ans. C

Sol. $S_1 = \frac{2n}{2}(2a + (2n-1)d) = n(2a + (2n-1)d)$

$$S_2 = \frac{4n}{2}(2a + (4n-1)d) = 2n(2a + (4n-1)d)$$

$$S_2 - S_1 = 1000$$

$$2n(2a + (4n-1)d) - n(2a + (2n-1)d) = 1000$$

$$n[4a + (4n-1)2d - 2a - (2n-1)d] = 1000$$

$$n[2a + d(8n-2-2n+1)] = 1000$$

$$n[2a + d(6n-1)] = 1000$$

$$\frac{6n}{2}(2a + (6n-1)d) = 3000$$

$$S = 3000$$

10. Let a_n be the n^{th} term of an A.P. If $\sum_{r=1}^{100} a_{2r} = \alpha$ and $\sum_{r=1}^{100} a_{2r-1} = \beta$, then the common difference of the A.P. is :

[AIEEE-20011]

माना एक समांतर श्रेणी का n वां पद a_n है। यदि $\sum_{r=1}^{100} a_{2r} = \alpha$ तथा $\sum_{r=1}^{100} a_{2r-1} = \beta$ है, तो समांतर श्रेणी का सार्व अन्तर है :

[AIEEE-20011]

- (A) $\frac{\alpha - \beta}{200}$ (B) $\alpha - \beta$ (C) $\frac{\alpha - \beta}{100}$ (D) $\beta - \alpha$

Ans. C

Sol. $\sum_{r=1}^{100} a_{2r} = \alpha$, $\sum_{r=1}^{100} a_{2r-1} = \beta$

$$a_2 + a_4 + a_6 + \dots + a_{200} = \alpha \dots (1)$$

$$\begin{aligned}
a_1 + a_3 + a_5 + \dots + a_{199} &= \beta \quad \dots(2) \\
(a_2 - a_1) + (a_4 - a_3) + \dots + (a_{200} - a_{199}) &= \alpha - \beta \\
d + d + \dots + d &= \alpha - \beta \\
(100 \text{ times}) &= \alpha - \beta \\
d &= \frac{\alpha - \beta}{100}
\end{aligned}$$

11. For any three positive real numbers a , b and c , $9(25a^2 + b^2) + 25(c^2 - 3ac) = 15b(3a + c)$. Then :

[JEE(Main)-2017]

- (A) a , b and c are in A.P. (B) a , b and c are in G.P.
 (C) b , c and a are in G.P. (D) b , c and a are in A.P.

किन्हीं तीन धनात्मक वास्तविक संख्याओं a , b तथा c के लिए $9(25a^2 + b^2) + 25(c^2 - 3ac) = 15b(3a + c)$ है, तो :

[JEE(Main)-2017]

- (A) a , b तथा c समान्तर श्रेणी है (B) a , b तथा c गुणोत्तर श्रेणी है
 (C) b , c तथा a गुणोत्तर श्रेणी है (D) b , c तथा a समान्तर श्रेणी है

Ans. D

Sol. $9(25a^2 + b^2) + 25(c^2 - 3ac) = 15b(3a + c)$
 $9((5a)^2 + b^2) + (5c)^2 - 3(5a)(5c) = 3(5a)(3b) + (3b)(5c)$
 $(15a)^2 + (3b)^2 + (5c)^2 - (15a)(5c) - (15a)(3b) - (3b)(5a) = 0$
 $(15a - 3b)^2 + (3b - 5c)^2 + (5c - 15a)^2 = 0$
 $15a = 3b = 5c = k$
 $a = \frac{k}{15}, b = \frac{k}{3}, c = \frac{k}{5}$
 $a = \left(\frac{k}{15}\right)45, b = \left(\frac{k}{3}\right)45, c = \left(\frac{k}{5}\right)45$
 $a = 3k, b = 15k, c = 9k$
 $2c = a + b$
 $\Rightarrow b, c, a$ are in A.P.

12. Let $a_1, a_2, a_3, \dots, a_{49}$ be in A.P. such that $\sum_{k=0}^{12} a_{4k+1} = 416$ and $a_9 + a_{43} = 66$. If $a_1^2 + a_2^2 + \dots + a_{17}^2 = 140m$,

then m is equal to :

[JEE(Main)-2018]

माना $a_1, a_2, a_3, \dots, a_{49}$ एक समांतर श्रेणी में ऐसे है कि $\sum_{k=0}^{12} a_{4k+1} = 416$ तथा $a_9 + a_{43} = 66$ है। यदि

$a_1^2 + a_2^2 + \dots + a_{17}^2 = 140m$ है, तो m बराबर है :

[JEE(Main)-2018]

- (A) 34 (B) 33 (C) 66 (D) 68

Ans. A

Sol. $a_1, a_2, a_3, \dots \rightarrow A.P.$

$$\sum_{k=0}^{12} a_{4k+1} = a_1 + a_5 + a_9 + \dots + a_{49} = 416$$

$$a_9 + a_{43} = 66$$

$$(a_1 + 8d) + (a_1 + 42d) = 66$$

$$2a_1 + 50d = 66$$

$$a_1 + 25d = 33 \quad \dots(1)$$

$$a_1 + (a_1 + 4d) + (a_1 + 8d) + (a_1 + 12d) + \dots + (a_1 + 48d) = 416$$

$$13a_1 + 4d[1 + 2 + 3 + \dots + 12] = 416$$

$$13a_1 + 4d \times \left(\frac{12 \times 13}{2} \right) = 416$$

$$13a_1 + 312d = 416$$

$$a_1 + 24d = 32 \quad \dots(2)$$

$$(1) - (2) : d = 1 \Rightarrow a_1 = 32 - 24 = 8 \Rightarrow a_1 = 8$$

Now,

$$a_1^2 + a_2^2 + \dots + a_{17}^2 = 140m$$

$$a_1^2 + (a_1^2 + d^2 + 2a_1d) + (a_1^2 + 4d^2 + 4a_1d) + \dots + (a_1^2 + 256d^2 + 32a_1d) = 140m$$

$$17a_1^2 + (2a_1d)[1 + 2 + 3 + \dots + 16] + d^2[1^2 + 2^2 + 3^2 + \dots + 16^2] = 140m$$

$$17a_1^2 + (2a_1d) \left[\frac{16 \times 17}{2} \right] + d^2 \left[\frac{16 \times 17 \times 33}{6} \right] = 140m$$

$$17 \times 64 + 16 \left[\frac{16 \times 17}{2} \right] + \left[\frac{16 \times 17 \times 33}{6} \right] = 140m$$

$$4760 = 140m \Rightarrow m = 34$$

GP & GM BASED

13. If $x > 1, y > 1, z > 1$ are in GP, then $\frac{1}{1 + \ln x}, \frac{1}{1 + \ln y}, \frac{1}{1 + \ln z}$ are in :

(A) AP

(B) HP

(C) GP

(D) None of these

यदि $x > 1, y > 1, z > 1$ गु.श्रे. में हों, तो $\frac{1}{1 + \ln x}, \frac{1}{1 + \ln y}, \frac{1}{1 + \ln z}$ किस श्रेणी में हैं :

(A) स.श्रे.

(B) ह.श्रे.

(C) गु.श्रे.

(D) इनमें से कोई नहीं

Ans. B

Sol. $x > 1, y > 1, z > 1$
 x, y, z are in G.P.
 $\Rightarrow \ln x, \ln y, \ln z$ are in A.P.
 $\Rightarrow 1 + \ln x, 1 + \ln y, 1 + \ln z$ are in A.P.
 $\Rightarrow \frac{1}{1 + \ln x}, \frac{1}{1 + \ln y}, \frac{1}{1 + \ln z}$ are in H.P.

14. For a sequence $\{a_n\}$, $a_1 = 2$ and $\frac{a_{n+1}}{a_n} = \frac{1}{3}$. Then $\sum_{r=1}^{20} a_r$ is :

- (A) $\frac{20}{2} [4 + 19 \times 3]$ (B) $3 \left(1 - \frac{1}{3^{20}}\right)$ (C) $2 (1 - 3^{20})$ (D) None of these

किसी अनुक्रम $\{a_n\}$ के लिए $a_1 = 2$ और $\frac{a_{n+1}}{a_n} = \frac{1}{3}$ हो, तो $\sum_{r=1}^{20} a_r$ का मान है :

- (A) $\frac{20}{2} [4 + 19 \times 3]$ (B) $3 \left(1 - \frac{1}{3^{20}}\right)$ (C) $2 (1 - 3^{20})$ (D) इनमें से कोई नहीं

Ans. B

Sol. $a_1 = 2$; $a_{n+1} = \frac{a_n}{3}$; $a_2 = \frac{a_1}{3} = \frac{2}{3}$; $a_3 = \frac{a_2}{3} = \frac{2}{3^2}$

$$a_1 + a_2 + \dots + a_{20} = 2 + \frac{2}{3} + \frac{2}{3^2} + \dots = \frac{2 \cdot \left[1 - \left(\frac{1}{3}\right)^{20}\right]}{1 - \frac{1}{3}} = 3 \left(1 - \frac{1}{3^{20}}\right)$$

15. α, β be the roots of the equation $x^2 - 3x + a = 0$ and γ, δ the roots of $x^2 - 12x + b = 0$ and numbers $\alpha, \beta, \gamma, \delta$ (in this order) form an increasing G.P., then :

α, β समीकरण $x^2 - 3x + a = 0$ के मूल एवं γ, δ समीकरण $x^2 - 12x + b = 0$ के मूल हैं और संख्याएँ $\alpha, \beta, \gamma, \delta$ (इसी क्रम में) वृद्धमान गु.श्रे. बनाती हैं, तो :

- (A) $a = 3, b = 12$ (B) $a = 12, b = 3$ (C) $a = 2, b = 32$ (D) $a = 4, b = 16$

Ans. C

Sol. $\alpha + \beta = 3$, $\alpha\beta = a$; $\gamma + \delta = 12$, $\gamma\delta = b$
 $\alpha, \beta, \gamma, \delta$ are in G.P. Let r be the common ratio so $\alpha(1+r) = 3$
 $\alpha r^2(1+r) = 12 \Rightarrow r^2 = 4 \Rightarrow r = 2$

Hindi. so $\alpha = 1 \Rightarrow$ so $a = 2$, $b = 32$ Ans
 $\alpha + \beta = 3$, $\alpha\beta = a$; $\gamma + \delta = 12$, $\gamma\delta = b$
 $\alpha, \beta, \gamma, \delta$ गुणोत्तर श्रेणी में हैं। माना कि r सार्वअनुपात है।
 अतः $\alpha(1+r) = 3 \Rightarrow \alpha r^2(1+r) = 12 \Rightarrow r^2 = 4 \Rightarrow r = 2$
 अतः $\alpha = 1$. अतः $a = 2$, $b = 32$ Ans

16. If the sum of n terms of a G.P. (with common ratio r) beginning with the p^{th} term is k times the sum of an equal number of terms of the same series beginning with the q^{th} term, then the value of k is :

यदि एक गु.श्रे. (सार्वअन्तर r) के p वें पद से प्रारम्भ होने वाले n पदों का योगफल, उसी श्रेणी के q वें पद से प्रारम्भ होने वाले समान पदों के योग का k गुना हो, तो k का मान है :

- (A) $r^{p/q}$ (B) $r^{q/p}$ (C) r^{p-q} (D) r^{p+q}

Ans. C

Sol. $\frac{ar^{p-1}(r^n - 1)}{r - 1} = k \cdot \frac{ar^{q-1}(r^n - 1)}{r - 1}$; $r^{p-1} = k \cdot r^{q-1}$; $k = r^{p-q}$

17. In an increasing geometric series, the sum of the second and the sixth term is $\frac{25}{2}$ and the product of the third and fifth term is 25. Then, the sum of 4^{th} , 6^{th} and 8^{th} terms is equal to : [JEE(Main)Feb.2021]

एक वर्धमान गुणोत्तर श्रेणी में, दूसरे तथा छठे पदों का योगफल $\frac{25}{2}$ है तीसरे तथा पाँचवें पदों का गुणनफल 25 है, तो चौथे, छठे तथा आठवें पदों का योगफल है :

[JEE(Main)Feb.2021]

- (A) 35 (B) 30 (C) 32 (D) 26

Ans. A

Sol. Let first term = a and common ratio = r .

Given $ar + ar^5 = \frac{25}{2}$ (1)

and $ar^2 \cdot ar^4 = 25 \Rightarrow ar^3 = 5$ (2)

From (1) & (2) ; $r = \sqrt{2}$ and $a = \frac{5}{2\sqrt{2}}$.

$t_4 + t_6 + t_8 = ar^3 + ar^5 + ar^7$
 $= ar^3 (1 + r^2 + r^4)$
 $= 5 (1 + 2 + 4) = 35$

18. In a geometric progression consisting of positive terms, each term equals the sum of the next two terms. Then the common ratio of this progression equals :

यदि किसी धनात्मक गुणोत्तर श्रेणी का प्रत्येक पद आने वाले पदों का योग हो, तो सार्वअन्तर होगा :

(A) $\frac{1}{2}(1-\sqrt{5})$ (B) $\frac{1}{2}\sqrt{5}$ (C) $\sqrt{5}$ (D) $\frac{1}{2}(\sqrt{5}-1)$

Ans. D

Sol. $a = ar + ar^2 \Rightarrow r^2 + r - 1 = 0 \Rightarrow r = \frac{-1 \pm \sqrt{5}}{2}$

$\Rightarrow r = \frac{\sqrt{5}-1}{2}$ (-ve not permissible)
(ऋणात्मक संभव नहीं है)

19. Three positive numbers form an increasing G.P. If the middle term in this G.P. is doubled, the new numbers are in A.P. Then the common ratio of the G.P. is : **[JEE(Main)-2014]**

तीन धनात्मक संख्याएँ एक वर्द्धमान गुणोत्तर श्रेणी बनाती है, यदि इसके मध्य को दुगना कर दिया जाये तो वे एक समान्तर श्रेणी में होंगे, तो गुणोत्तर श्रेणी का सार्वअन्तर होगा : **[JEE(Main)-2014]**

(A) $2-\sqrt{3}$ (B) $2+\sqrt{3}$ (C) $\sqrt{2}+\sqrt{3}$ (D) $3+\sqrt{2}$

Ans. B

Sol. $\frac{a}{r}, a, ar$ are in G.P.

$\frac{a}{r}, 2a, ar$ are in A.P.

$2(2a) = \frac{a}{r} + ar$

$4 = \frac{1}{r} + r$

$r^2 + 1 = 4r$

$r^2 - 4r + 1 = 0$

$r = \frac{4 \pm \sqrt{16-4}}{2} = 2 \pm \sqrt{3}$

$\therefore$ Our sequence is in increasing G.P., so $r > 1$

$r = 2 + \sqrt{3}$

20. If m is the A.M. of two distinct real numbers l and n ($l, n > 1$) and G_1, G_2 and G_3 are three geometric means between l and n , then $G_1^4 + 2G_2^4 + G_3^4$ equals : **[JEE(Main)-2015]**

यदि दो विभिन्न वास्तविक संख्याओं l तथा n ($l, n > 1$) का समांतर माध्य (A.M.) m है और l तथा n के बीच तीन गुणोत्तर माध्य (G.M.) G_1, G_2 तथा G_3 हैं, तो $G_1^4 + 2G_2^4 + G_3^4$ बराबर है : **[JEE(Main)-2015]**

- (A) $4l^2mn$ (B) $4lm^2n$ (C) $4lmn^2$ (D) $4l^2m^2n^2$

Ans. B

Sol. $m = \frac{n + l}{2} \dots(1)$

l, G, G_2, G_3, n are in G.P.

$$\Rightarrow G_1 = \ell r, G_2 = \ell r^2, G_3 = \ell r^3, n = \ell r^4$$

$$n = \ell r^4 \Rightarrow r = \left(\frac{n}{\ell}\right)^{\frac{1}{4}}$$

$$\begin{aligned} \text{Now } G_1^4 + 2G_2^4 + G_3^4 &= \left[\ell \left(\frac{n}{\ell}\right)^{\frac{1}{4}}\right]^4 + 2\left[\ell^4 \left(\frac{n}{\ell}\right)^{\frac{2}{4}}\right]^4 + \left[\ell \left(\frac{n}{\ell}\right)^{\frac{3}{4}}\right]^4 \\ &= \ell^4 \left(\frac{n}{\ell}\right) + 2\left[\ell^4 \left(\frac{n}{\ell}\right)^2\right] + \left[\ell^4 \left(\frac{n}{\ell}\right)^3\right] \\ &= n\ell^3 + 2\ell^2 n^2 + \ell n^3 \\ &= n\ell(\ell^2 + 2\ell n + n^2) \\ &= n\ell(\ell + n)^2 \\ &= n\ell(2m)^2 \quad (\text{By equation (1)}) \\ &= 4m^2 n\ell \\ &= 4\ell m^2 n \end{aligned}$$

21. If the 2nd, 5th and 9th terms of a non-constant A.P. are in G.P., then the common ratio of this G.P. is : **[JEE(Main)-2016]**

यदि एक अचरेतर समान्तर श्रेढ़ी का 2nd, 5th तथा 9th पद एक गुणोत्तर श्रेढ़ी में है, तो उस गुणोत्तर श्रेढ़ी का सार्व अनुपात है : **[JEE(Main)-2016]**

(A) 1

(B) $7/4$ (C) $8/5$ (D) $4/3$ **Ans.** D

Sol. T_2, T_5, T_9 are in G.P.
 $(A + d), (a + 4d), (a + 8d)$ are in G.P.

$$r = \frac{a + 4d}{a + d} = \frac{a + 8d}{a + 4d}$$

$$\Rightarrow (a + 4d)^2 = (a + d)(a + 8d)$$

$$a^2 + 16d^2 + 8ad = a^2 + 8d^2 + 9ad$$

$$8d^2 - ad = 0$$

$$d(8d - a) = 0$$

$$d \neq 0 \Rightarrow a = 8d$$

$$\text{Now } \Rightarrow r = \frac{a + 4d}{a + d} = \frac{12d}{9d} = \frac{4}{3}$$

AM GM HM

22. If A, G and H are respectively the A.M., G.M. & H.M. of three positive numbers a, b, and c, then the equation whose roots are a, b, and c is given by :

यदि तीन धनात्मक संख्याओं a, b, c का समान्तर माध्य, गुणोत्तर माध्य एवं हरात्मक माध्य क्रमशः A, G एवं H हो, तो वह समीकरण जिसके मूल a, b, c हैं, होगी :

$$(A) \quad x^3 - 3Ax^2 + 3G^3x - G^3 = 0$$

$$(B) \quad x^3 - 3Ax^2 + 3(G^3/H)x - G^3 = 0$$

$$(C) \quad x^3 + 3Ax^2 + 3(G^3/H)x - G^3 = 0$$

$$(D) \quad x^3 - 3Ax^2 - 3(G^3/H)x + G^3 = 0$$

Ans. B

$$\text{Sol. } AM = A = \frac{a + b + c}{3} \quad ; \quad GM = G = (abc)^{1/3}$$

$$HM = H = \frac{3abc}{ab + bc + ca} = \frac{3G^3}{ab + bc + ca}$$

$$\text{Equation whose roots are a, b, c } \Rightarrow x^3 - (a + b + c)x^2 + (\Sigma ab)x - abc = 0$$

$$\Rightarrow x^3 - 3Ax^2 + \frac{3G^3}{H}x - G^3 = 0 \quad \text{Ans}$$

$$\text{Hindi. समान्तर माध्य} = A = \frac{a + b + c}{3}; \quad \text{गुणोत्तर माध्य} = G = (abc)^{1/3}$$

$$\text{हरात्मक माध्य} = H = \frac{3abc}{ab + bc + ca} = \frac{3G^3}{ab + bc + ca}$$

समीकरण जिसके मूल a, b, c हो, निम्न है -

$$\Rightarrow x^3 - (a + b + c)x^2 + (\Sigma ab)x - abc = 0 \quad \Rightarrow x^3 - 3Ax^2 + \frac{3G^3}{H}x - G^3 = 0 \quad \text{Ans}$$

23. If $x_i > 0, i = 1, 2, \dots, 50$ and $x_1 + x_2 + \dots + x_{50} = 50$, then the minimum value of $\frac{1}{x_1} + \frac{1}{x_2} + \dots + \frac{1}{x_{50}}$ equal to :

यदि $x_i > 0, i = 1, 2, \dots, 50$ और $x_1 + x_2 + \dots + x_{50} = 50$ हो, तो $\frac{1}{x_1} + \frac{1}{x_2} + \dots + \frac{1}{x_{50}}$ का न्यूनतम मान है :

- (A) 50 (B) $(50)^2$ (C) $(50)^3$ (D) $(50)^4$

Ans. A

Sol. $x_1 + x_2 + x_3 + \dots + x_{50} = 50$
AM $\geq$ HM

$$\frac{x_1 + x_2 + \dots + x_{50}}{50} \geq \frac{1}{\frac{\frac{1}{x_1} + \frac{1}{x_2} + \frac{1}{x_3} + \dots + \frac{1}{x_{50}}}{50}}$$

$$\Rightarrow \frac{x_1 + x_2 + \dots + x_{50}}{50} \geq \frac{50}{\frac{1}{x_1} + \frac{1}{x_2} + \dots + \frac{1}{x_{50}}} \Rightarrow \frac{1}{x_1} + \frac{1}{x_2} + \dots + \frac{1}{x_{50}} \geq 50$$

so min value of $\frac{1}{x_1} + \frac{1}{x_2} + \dots + \frac{1}{x_{50}} = 50$ अतः $\frac{1}{x_1} + \frac{1}{x_2} + \dots + \frac{1}{x_{50}}$ का न्यूनतम मान = 50

24. If A_1, A_2 be two A.M.s and G_1, G_2 be two G.M.s between a and b , then $\frac{A_1 + A_2}{G_1 G_2}$ is equal to :

यदि a और b के मध्य दो समान्तर माध्य A_1, A_2 और दो गुणोत्तर माध्य G_1, G_2 हो, तो $\frac{A_1 + A_2}{G_1 G_2}$ का मान है :

- (A) $\frac{a+b}{2ab}$ (B) $\frac{2ab}{a+b}$ (C) $\frac{a+b}{ab}$ (D) $\frac{a+b}{\sqrt{ab}}$

Ans. C

Sol. a, A_1, A_2, b are in A.P.

$$A_1 = a + d, A_2 = a + 2d, \quad b = a + 3d \Rightarrow \frac{b-a}{3} = d$$

$$A_1 + A_2 = 2a + 3d = 2a + 3\left(\frac{b-a}{3}\right) \\ = 2a + b - a = a + b$$

a, G_1, G_2, b are in G.P.
 $G_1 = ar, G_2 = ar^2, b = ar^3$
 $G_1 G_2 = a^2 r^3 = ab$

Now $\frac{A_1 + A_2}{G_1 G_2} = \frac{a + b}{ab}$

25. If P, Q be the A.M., G.M. respectively between any two rational numbers a and b , then $P - Q$ is equal to :

किन्हीं दो परिमेय संख्याओं a एवं b के मध्य गुणोत्तर माध्य एवं समान्तर माध्य क्रमशः P एवं Q हो, तो $P - Q$ का मान है :

(A) $\frac{a-b}{a}$ (B) $\frac{a+b}{2}$ (C) $\frac{2ab}{a+b}$ (D) $\left(\frac{\sqrt{a}-\sqrt{b}}{\sqrt{2}}\right)^2$

Ans. D

Sol. $P = \frac{a+b}{2}, Q = \sqrt{ab}$; $P - Q = \frac{a+b-2\sqrt{ab}}{2} = \frac{(\sqrt{a}-\sqrt{b})^2}{2}$

26. If $\alpha \in \left(0, \frac{\pi}{2}\right)$ then $\sqrt{x^2 + x} + \frac{\tan^2 \alpha}{\sqrt{x^2 + x}}$ is always greater than or equal to :

यदि $\alpha \in \left(0, \frac{\pi}{2}\right)$ हो, तो $\sqrt{x^2 + x} + \frac{\tan^2 \alpha}{\sqrt{x^2 + x}}$ का न्यूनतम मान है :

(A) $2 \tan \alpha$ (B) 1 (C) 2 (D) $\sec^2 \alpha$

Ans. A

Sol. $\sqrt{x^2 + x} + \frac{\tan^2 \alpha}{\sqrt{x^2 + x}} \geq 2 \sqrt{\left(\sqrt{x^2 + x}\right) \times \left(\frac{\tan^2 \alpha}{\sqrt{x^2 + x}}\right)} = 2 \tan \alpha$

HP & HM BASED

27. The harmonic mean of the roots of the equation $(5 + \sqrt{2})x^2 - (4 + \sqrt{5})x + 8 + 2\sqrt{5} = 0$ is :

समीकरण $(5 + \sqrt{2})x^2 - (4 + \sqrt{5})x + 8 + 2\sqrt{5} = 0$ के मूलों का हरात्मक माध्य है :

(A) 2 (B) 4 (C) 6 (D) 8

Ans. B

Sol. $(5 + \sqrt{2})x^2 - (4 + \sqrt{5})x + (8 + 2\sqrt{5}) = 0$

$$\alpha + \beta = \frac{-(4 + \sqrt{5})}{(5 + \sqrt{2})} \quad \alpha\beta = \frac{8 + 2\sqrt{5}}{5 + \sqrt{2}}$$

$$H = \frac{2\alpha\beta}{\alpha + \beta} = \frac{2(8 + 2\sqrt{5})}{4 + \sqrt{5}} = 4$$

28. If $a_1, a_2, \dots, a_n$ are in H.P., then the expression $a_1a_2 + a_2a_3 + \dots + a_{n-1}a_n$ is equal to :

यदि $a_1, a_2, \dots, a_n$ हरात्मक श्रेणी में है, तो व्यंजक $a_1a_2 + a_2a_3 + \dots + a_{n-1}a_n$ का मान है :

(A) $(n-1)(a_1 - a_n)$ (B) na_1a_n (C) $(n-1)a_1a_n$ (D) $n(a_1 - a_n)$

Ans. C

Sol. $\frac{1}{a_1}, \frac{1}{a_2}, \dots, \frac{1}{a_n}$ are in A.P. समान्तर श्रेणी में है।

$$\Rightarrow \frac{1}{a_2} - \frac{1}{a_1} = \dots = \frac{1}{a_n} - \frac{1}{a_{n-1}} = d$$

$$\Rightarrow a_1a_2 = \frac{a_1 - a_2}{d}, a_2a_3 = \frac{a_2 - a_3}{d}, \dots, \frac{a_{n-1} - a_n}{d} = a_{n-1} - a_n$$

$$\Rightarrow a_1a_2 + a_2a_3 + \dots + a_{n-1}a_n = \frac{a_1 - a_2}{d} + \dots + \frac{a_{n-1} - a_n}{d} = \frac{a_1 - a_n}{d} = (n-1)a_1a_n$$

29. If $x = \sum_{n=0}^{\infty} a^n, y = \sum_{n=0}^{\infty} b^n, z = \sum_{n=0}^{\infty} c^n$ where a, b, c are in A.P. and $|a| < 1, |b| < 1, |c| < 1$ then x, y, z are in :

(A) GP (B) AP
(C) Arithmetic - Geometric Progression (D) HP

यदि $x = \sum_{n=0}^{\infty} a^n, y = \sum_{n=0}^{\infty} b^n, z = \sum_{n=0}^{\infty} c^n$ जबकि a, b, c समान्तर श्रेणी में है तथा $|a| < 1, |b| < 1$ तथा $|c| < 1$ तो x, y तथा z :

(A) गुणोत्तर श्रेणी में है (B) समान्तर श्रेणी में है
(C) समान्तर गुणोत्तर श्रेणी में है (D) हरात्मक श्रेणी में है

Ans. D

Sol. $x = \frac{1}{1-a}, y = \frac{1}{1-b}, z = \frac{1}{1-c} \Rightarrow a, b, c$ are in A.P. समान्तर श्रेणी में है।

$\Rightarrow 1-a, 1-b, 1-c$ are also in A.P. भी समान्तर श्रेणी में है।

$\Rightarrow \frac{1}{1-a}, \frac{1}{1-b}, \frac{1}{1-c}$ are in H.P. हरात्मक श्रेणी में है।

AGP

30. The sum of infinite series $1 - \frac{3}{2} + \frac{5}{4} - \frac{7}{8} + \dots$ is :

श्रेणी $1 - \frac{3}{2} + \frac{5}{4} - \frac{7}{8} + \dots$ के अनन्त पदों का योगफल है :

(A) $2/9$ (B) $2/3$ (C) $-2/9$ (D) $9/2$

Ans. A

Sol. $S = 1 - \frac{3}{2} + \frac{5}{4} - \frac{7}{8} + \dots$

$$\frac{S}{2} = \frac{1}{2} - \frac{3}{4} + \frac{5}{8} - \dots$$

$$\frac{3S}{2} = 1 - \frac{2}{2} + \frac{2}{4} - \frac{2}{8} + \dots$$

$$= 1 - 2 \left(\frac{\frac{1}{2}}{1 + \frac{1}{2}} \right)$$

$$\frac{3S}{2} = 1 - 2 \left(\frac{1}{3} \right) = \frac{+1}{3}$$

$$S = \frac{2}{9}$$

31. If $(10)^9 + 2(11)^1 (10)^8 + 3(11)^2 (10)^7 + \dots + 10(11)^9 = k(10)^9$, then k is equal to : **[JEE(Main)-2014]**

यदि $(10)^9 + 2(11)^1 (10)^8 + 3(11)^2 (10)^7 + \dots + 10(11)^9 = k(10)^9$ है, तो k बराबर है : **[JEE(Main)-2014]**

(A) 100 (B) 110 (C) $121/10$ (D) $441/100$

Ans. A

Sol. $(10)^9 + 2(11)^1 (10)^8 + 3(11)^2 (10)^7 + \dots + 10(11)^9 = k(10)^9$

$$1 + 2\left(\frac{11}{10}\right) + 3\left(\frac{11}{10}\right)^2 + \dots + 10\left(\frac{11}{10}\right)^9 = k \quad \dots(1)$$

$$0 + \left(\frac{11}{10}\right) + 2\left(\frac{11}{10}\right)^2 + \dots + 9\left(\frac{11}{10}\right)^9 + 10\left(\frac{11}{10}\right)^{10} = \frac{11k}{10} \quad \dots(2)$$

$$(2) - (1)$$

$$\frac{k}{10} = 10\left(\frac{11}{10}\right)^{10} - \left[1 + \left(\frac{11}{10}\right) + \left(\frac{11}{10}\right)^2 + \dots + \left(\frac{11}{10}\right)^9\right]$$

$$\frac{k}{10} = 10\left(\frac{11}{10}\right)^{10} - \left[1 \cdot \frac{\left(\left(\frac{11}{10}\right)^{10} - 1\right)}{\frac{11}{10} - 1}\right]$$

$$\frac{k}{10} = 10\left(\frac{11}{10}\right)^{10} - \left[10\left(\frac{11}{10}\right)^{10} - 10\right]$$

$$\frac{k}{10} = 10$$

$$k = 100$$

MISCELLANEOUS SEQUENCES

32. The sum of the series $1^3 - 2^3 + 3^3 - \dots + 9^3 =$

श्रेणी $1^3 - 2^3 + 3^3 - \dots + 9^3$ का योग है :

- (A) 300 (B) 125 (C) 425 (D) 0

Ans. C

Sol. $S = 1^3 - 2^3 + 3^3 - 4^3 + \dots + 9^3$

$$r = 1$$

$$S = (1^3 + 3^3 + 5^3 + 7^3 + 9^3) - (2^3 + 4^3 + 6^3 + 8^3)$$

$$= (1^3 + 3^3 + 5^3 + 7^3 + 9^3) - 8(1^3 + 2^3 + 3^3 + 4^3)$$

$$= (1^3 + 2^3 + 3^3 + 4^3 + 5^3 + 6^3 + 7^3 + 8^3 + 9^3) - (2^3 + 4^3 + 6^3 + 8^3) - 8(1^3 + 2^3 + 3^3 + 4^3)$$

$$= \left[\frac{9(10)}{2}\right]^2 - 16\left[\frac{4 \times 5}{2}\right]^2 = 425$$

33. If $\frac{1}{1^2} + \frac{1}{2^2} + \frac{1}{3^2} + \dots$ upto $\infty = \frac{\pi^2}{6}$, then $\frac{1}{1^2} + \frac{1}{3^2} + \frac{1}{5^2} + \dots =$

- (A) $\pi^2/12$ (B) $\pi^2/24$ (C) $\pi^2/8$ (D) None of these

यदि $\frac{1}{1^2} + \frac{1}{2^2} + \frac{1}{3^2} + \dots = \frac{\pi^2}{6}$ हो, तो $\frac{1}{1^2} + \frac{1}{3^2} + \frac{1}{5^2} + \dots =$

- (A) $\pi^2/12$ (B) $\pi^2/24$ (C) $\pi^2/8$ (D) इनमें से कोई नहीं

Ans. C

Sol. $\therefore I = \frac{1}{1^2} + \frac{1}{2^2} + \frac{1}{3^2} + \dots \infty = \frac{\pi^2}{6}$

Let $\frac{1}{1^2} + \frac{1}{3^2} + \frac{1}{5^2} + \dots = A \quad \therefore I = \left(\frac{1}{1^2} + \frac{1}{3^2} + \frac{1}{5^2} + \dots \infty \right) + \frac{1}{2^2} \left(\frac{1}{1^2} + \frac{1}{2^2} + \frac{1}{3^2} + \dots \infty \right)$

$\Rightarrow I = A + \frac{I}{4} \Rightarrow A = \frac{3I}{4} = \frac{3}{4} \times \frac{\pi^2}{6} \Rightarrow A = \frac{\pi^2}{8}$

Hindi $\therefore I = \frac{1}{1^2} + \frac{1}{2^2} + \frac{1}{3^2} + \dots \infty = \frac{\pi^2}{6}$

माना $\frac{1}{1^2} + \frac{1}{3^2} + \frac{1}{5^2} + \dots = A \quad \therefore I = \left(\frac{1}{1^2} + \frac{1}{3^2} + \frac{1}{5^2} + \dots \infty \right) + \frac{1}{2^2} \left(\frac{1}{1^2} + \frac{1}{2^2} + \frac{1}{3^2} + \dots \infty \right)$

$\Rightarrow I = A + \frac{I}{4} \Rightarrow A = \frac{3I}{4} = \frac{3}{4} \times \frac{\pi^2}{6} \Rightarrow A = \frac{\pi^2}{8}$

34. The sum of first 9 terms of the series $\frac{1^3}{1} + \frac{1^3 + 2^3}{1+3} + \frac{1^3 + 2^3 + 3^3}{1+3+5} + \dots$ is : **[JEE(Main)2015]**

श्रेणी $\frac{1^3}{1} + \frac{1^3 + 2^3}{1+3} + \frac{1^3 + 2^3 + 3^3}{1+3+5} + \dots$ के प्रथम 9 पदों का योग है : **[JEE(Main)-2015]**

- (A) 71 (B) 96 (C) 142 (D) 192

Ans. B

Sol. $T_n = \frac{1^3 + 2^3 + 3^3 + \dots + n^3}{1 + 3 + 5 + \dots + (2n-1)} = \frac{\left[\frac{n(n+1)}{2} \right]^2}{n^2}$

$T_n = \left(\frac{n+1}{2} \right)^2$

$S_9 = T_1 + T_2 + T_3 + \dots + T_9$

$$\begin{aligned}
&= \frac{1}{(2)^2} \left[(2)^2 + (3)^2 + (4)^2 + \dots + (10)^2 \right] \\
&= \frac{1}{4} \left[\left[(1)^2 + (2)^2 + (3)^2 + \dots + (10)^2 \right] - 1 \right] \\
&= \frac{1}{4} \left[\left\{ \frac{10 \times 11 \times 21}{6} \right\} - 1 \right] = 96
\end{aligned}$$

35. If the sum of the first ten terms of the series $\left(1\frac{3}{5}\right)^2 + \left(2\frac{2}{5}\right)^2 + \left(3\frac{1}{5}\right)^2 + 4^2 + \left(4\frac{4}{5}\right)^2 + \dots$, is $\frac{16}{5}m$, then m is :

[JEE(Main)-2016]

यदि श्रेणी $\left(1\frac{3}{5}\right)^2 + \left(2\frac{2}{5}\right)^2 + \left(3\frac{1}{5}\right)^2 + 4^2 + \left(4\frac{4}{5}\right)^2 + \dots$, के प्रथम दस पदों का योग $\frac{16}{5}m$ है, तो m बराबर है :

[JEE(Main)-2016]

- (A) 100 (B) 99 (C) 102 (D) 101

Ans. D

Sol. $\left(1\frac{3}{5}\right)^2 + \left(2\frac{2}{5}\right)^2 + \left(3\frac{1}{5}\right)^2 + 4^2 + \left(4\frac{4}{5}\right)^2 + \dots$

$$\begin{aligned}
&\left(\frac{8}{5}\right)^2 + \left(\frac{12}{5}\right)^2 + \left(\frac{16}{5}\right)^2 + (4)^2 + \left(\frac{24}{5}\right)^2 + \dots \\
&\left(\frac{8}{5}\right)^2 + \left(\frac{12}{5}\right)^2 + \left(\frac{16}{5}\right)^2 + \left(\frac{20}{5}\right)^2 + \left(\frac{24}{5}\right)^2 + \dots \\
&\left(\frac{8}{5}\right)^2 + \left(\frac{12}{5}\right)^2 + \dots + \left(\frac{44}{5}\right)^2 = \frac{16}{5}m \\
&\left(\frac{4}{5}\right)^2 \left[(2)^2 + (3)^2 + \dots + (11)^2 \right] = \frac{16}{5}m \\
&\frac{1}{5} \left[\frac{11 \times 12 \times 23}{6} - 1 \right] = m \Rightarrow m = 101
\end{aligned}$$

36. If, for a positive integer n, the quadratic equation, $x(x+1) + (x+1)(x+2) + \dots + (x+n-1)(x+n) = 10n$ has two consecutive integral solutions, then n is equal to :

[JEE(Main)2017]

यदि किसी घनपूर्णांक n के लिए, द्विघाती समीकरण $x(x+1) + (x+1)(x+2) + \dots + (x+n-1)(x+n) = 10n$ के दो क्रमित पूर्णांकीय हल हैं, तो n बराबर है : **[JEE(Main)-2017]**

- (A) 10 (B) 11 (C) 12 (D) 9

Ans. B

Sol.
$$\sum_{r=1}^n (x + (r-1))(x + r) = 10n$$

$$\sum_{r=1}^n [x^2 + x(2r-1) + (r^2 - r)] = 10n$$

$$(x^2 - x)n + n(n+1)x + \frac{n(n+1)(2n+1)}{6} - \frac{n(n+1)}{2} = 10n$$

$$10 = x^2 - x + 1(n+1)x + \left(\frac{n^2 - 1}{3}\right)$$

$$x^2 + nx + \left(\frac{n^2 - 31}{3}\right) = 0 \quad \alpha, \alpha + 1 \text{ are roots of this quadratic equation.}$$

$$2\alpha + 1 = -n \quad \Rightarrow \alpha = -\left(\frac{n+1}{2}\right)$$

$$\alpha(\alpha + 1) = \frac{n^2 - 31}{3}$$

$$-\left(\frac{n+1}{2}\right)\left(\frac{1-n}{2}\right) = \frac{n^2 - 31}{3}$$

$$3(n^2 - 1) = 4n^2 - 124$$

$$n^2 - 121 = 0$$

$$n = \pm 11$$

$$\Rightarrow n = 11$$

37. Let A be the sum of the first 20 terms and B be sum of the first 40 terms of the series $1^2 + 2 \cdot 2^2 + 3^2 + 2 \cdot 4^2 + 5^2 + 2 \cdot 6^2 + \dots$. If $B - 2A = 100\lambda$, then λ is equal to : **[JEE(Main)-2018]**

माना श्रेणी $1^2 + 2 \cdot 2^2 + 3^2 + 2 \cdot 4^2 + 5^2 + 2 \cdot 6^2 + \dots$ के प्रथम 20 पदों का योग A है तथा प्रथम 40 पदों का योग B है। यदि $B - 2A = 100\lambda$, तो λ बराबर है : **[JEE(Main)-2018]**

- (A) 464 (B) 496 (C) 232 (D) 248

Ans. D

Sol. $1^2 + 2 \cdot 2^2 + 3^2 + 2 \cdot 4^2 + 5^2 + 2 \cdot 6^2 + \dots$

$$S_{20} = A, S_{40} = B$$

$$\begin{aligned} A = S_{20} &= 1^2 + 2 \cdot 2^2 + 3^2 + \dots + 19^2 + 2 \cdot (20)^2 \\ &= [1^2 + 3^2 + 5^2 + \dots + 19^2] + [2 \cdot 2^2 + 2 \cdot 4^2 + \dots + 2 \cdot (20)^2] \\ &= [1^2 + 3^2 + 5^2 + \dots + 19^2] + 2 \cdot 2^2 [1 + 2^2 + 3^2 + 10^2] \\ &= [1^2 + 2^2 + 3^2 + \dots + 19^2 + 20^2] + 8[1^2 + 2^2 + \dots + 10^2] - [2^2 + 4^2 + 6^2 + \dots + 20^2] \end{aligned}$$

$$A = \left(\frac{20 \times 21 \times 41}{6} \right) + 4 \left(\frac{10 \times 11 \times 21}{6} \right)$$

$$\begin{aligned} B &= 1^2 + 2 \cdot 2^2 + \dots + (39)^2 + (40)^2 \\ &= [1^2 + 2^2 + 3^2 + \dots + (40)^2] + 2 \cdot 2^2 [1 + 2^2 + 3^2 + \dots + (20)^2] - 4[1 + 2^2 + \dots + (20)^2] \end{aligned}$$

$$B = \left(\frac{40 \times 41 \times 81}{6} \right) + 4 \left(\frac{20 \times 21 \times 41}{6} \right)$$

$$\begin{aligned} B - 2A &= \left(\frac{40 \times 41 \times 81}{6} \right) + 4 \left(\frac{20 \times 21 \times 41}{6} \right) - 2 \left(\frac{20 \times 21 \times 41}{6} \right) - 8 \left(\frac{10 \times 11 \times 21}{6} \right) \\ &= 24800 = 100\lambda \Rightarrow \lambda = 248 \end{aligned}$$

38. The coefficient of x^8 in the polynomial $(x-1)(x-2)(x-3) \dots (x-10)$ is :

बहुपद $(x-1)(x-2)(x-3) \dots (x-10)$ में x^8 का गुणांक होगा :

- (A) 2640 (B) 1320 (C) 1370 (D) 2740

Ans. B

Sol. $f(x) = (x-1)(x-2)(x-3) \dots (x-9)(x-10)$
 $x^{10} - [1 + 2 + \dots + 10]x^9 + (1 \cdot 2 + 1 \cdot 3 + 1 \cdot 4 + \dots) x^8 + \dots$
 Coefficient of $x^8 = (1 \cdot 2 + 1 \cdot 3 + \dots + 1 \cdot 10) + (2 \cdot 3 + \dots + 2 \cdot 10) + (3 \cdot 4 + \dots + 3 \cdot 10) + \dots + (9 \cdot 10)$
 $(1 + 2 + 3 + \dots + 10)^2 = (1^2 + 2^2 + \dots + 10^2) + 2[(1 \cdot 2 + \dots + 1 \cdot 10) + \dots + (9 \cdot 10)]$
 $\left(\frac{10 \times 11}{2} \right)^2 = \frac{10 \times 11 \times 21}{6} + 2[\text{Coefficient of } x^8]$
 $\frac{2640}{2} = \text{coefficient of } x^8$
 Coeff. of $x^8 = 1320$

39. Find the value of $\frac{2}{1^3} + \frac{6}{1^3 + 2^3} + \frac{12}{1^3 + 2^3 + 3^3} + \frac{20}{1^3 + 2^3 + 3^3 + 4^3} + \dots$ upto ∞ terms :

$\frac{2}{1^3} + \frac{6}{1^3 + 2^3} + \frac{12}{1^3 + 2^3 + 3^3} + \frac{20}{1^3 + 2^3 + 3^3 + 4^3} + \dots$ से ∞ पदों तक का मान होगा :

- (A) 2 (B) $\frac{1}{2}$ (C) 4 (D) $\frac{1}{4}$

Ans. C

Sol. $S = \frac{2}{1^3} + \frac{6}{1^3 + 2^3} + \frac{12}{1^3 + 2^3 + 3^3} + \frac{20}{1^3 + 2^3 + 3^3 + 4^3} + \dots$

$$T_n = \frac{n(n+1)}{\left(\frac{n(n+1)}{2}\right)^2} = \frac{4}{n(n+1)}$$

$$S = \sum_{n=1}^{\infty} \frac{4}{n(n+1)} = 4 \sum_{n=1}^{\infty} \frac{(n+1) - n}{n(n+1)}$$

$$S = 4 \sum_{n=1}^{\infty} \left[\frac{1}{n} - \frac{1}{n+1} \right] = 4$$